Reproducing kernel The reproducing kernel of $\mathcal{H}_n^d(w_\kappa)$ is given by

$$P_n(w_\kappa; x, y) = \sum_{1 \le j \le \dim \mathcal{H}_n^d(w_\kappa)} Y_j(x) Y_j(y)$$

where $\{Y_j\}_{j=1}^{\dim \mathcal{H}_n^d(w_\kappa)}$ is an orthonormal basis of $\mathcal{H}_n^d(w_\kappa)$. It satisfies a closed formula

$$P_n(w_\kappa; x, y) = \frac{n + |\kappa| + \frac{1}{2}d - 1}{|\kappa| + \frac{1}{2}d - 1} \int_{[-1,1]^d} C_n^{|\kappa|+\frac{1}{2}d-1}(x_1 y_1 t_1 + \cdots + x_d y_d t_d) \times \left(\prod_{i=1}^{d} c_{\kappa_i} (1 + t_i)(1 - t_i^2)^{\kappa_i - 1} \right) dt. \quad (2.4.24)$$

Further results and references The h-harmonics associated with a finite reflection group were first studied by Dunkl in [26]. Next he defined his Dunkl operators in [27]. For an overview of the extensive theory of h-harmonics, see [28] and Chapter 7. The case $\mathbb{Z}_2^d$ was studied in detail in [119], which contains (2.4.23) and (2.4.24), as well as a closed formula for an analog of the Poisson integral. A Funk–Hecke type formula was given in [123]. For a monic h-harmonic basis and a biorthogonal basis, see [128]. For a connection to products of Heine–Stieltjes polynomials, see [110].

2.5 Classical orthogonal polynomials of several variables

General properties of orthogonal polynomials of several variables were given in §2.2. This section contains results for specific weight functions.

2.5.1 Classical orthogonal polynomials on the unit ball

On the unit ball $B^d := \{x \in \mathbb{R}^d :| \, \|x\| \le 1\}$, consider the weight function

$$W_\mu(x, y) := \frac{\Gamma(\mu + \frac{1}{2}(d+1))}{\pi^{d/2}\Gamma(\mu + \frac{1}{2})} (1 - \|x\|^2)^{\mu - \frac{1}{2}}, \quad \mu > -\tfrac{1}{2}, \quad (2.5.1)$$

normalized such that its integral over B^d is 1. For $d = 2$, see §2.3.2.

Differential operator Orthogonal polynomials of degree n with respect to W_μ are eigenfunctions of a second order differential operator:

$$\left(\Delta - \sum_{j=1}^{d} \frac{\partial}{\partial x_j} x_j \left((2\mu - 1) + \sum_{i=1}^{d} x_i \frac{\partial}{\partial x_i} \right) \right) P = -(n + d)(n + 2\mu - 1)P, \quad P \in \mathcal{V}_n^d, \quad (2.5.2)$$

where Δ is the Laplace operator.

First orthonormal basis Associated with $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, define $\mathbf{x}_j := (x_1, \dots, x_j)$ for $1 \le j \le d$ and $\mathbf{x}_0 := 0$. For $\alpha \in \mathbb{N}_0^d$ and $1 \le j \le d$, let $\alpha^j := (\alpha_j, \dots, \alpha_d)$ and $\alpha^{d+1} := 0$. An orthonormal basis $\{P_\alpha\}_{|\alpha|=n}$ of $\mathcal{V}_n^d$ is given by

$$P_\alpha(x) := [h_\alpha]^{-1} \prod_{j=1}^{d} (1 - \|\mathbf{x}_{j-1}\|^2)^{\frac{1}{2}\alpha_j} C_{\alpha_j}^{\lambda_j}\Big(\frac{x_j}{(1 - \|\mathbf{x}_{j-1}\|^2)^{1/2}}\Big), \tag{2.5.3}$$

where $\lambda_j := \mu + |\alpha^{j+1}| + \frac{1}{2}(d - j)$ and h_α is given by

$$[h_\alpha]^2 := \frac{(\mu + \frac{1}{2}d)_{|\alpha|}}{(\mu + \frac{1}{2}(d+1))_{|\alpha|}} \prod_{j=1}^{d} \frac{(\mu + \frac{1}{2}(d-j))_{|\alpha^j|} (2\mu + 2|\alpha^{j+1}| + d - j)_{\alpha_j}}{(\mu + \frac{1}{2}(d - j + 1))_{|\alpha^j|} \alpha_j!}, \tag{2.5.4}$$

and where the case $\mu = 0$ can be obtained as a limit for $\mu \to 0$ by using (2.3.1), similarly as for (2.3.6), (2.3.7).

Second orthonormal basis Let $r_k^d := \dim \mathcal{H}_k^d$. For $0 \le j \le n/2$ let $\{Y_{\ell,n-2j} \mid 1 \le \ell \le r_{n-2j}^d\}$ be an orthonormal basis of $\mathcal{H}_{n-2j}^d$, the space of spherical harmonics of degree $n - 2j$, with respect to the normalized surface measure. For $0 \le j \le n/2$, define

$$P_{\ell,j}(x) := [h_{j,n}]^{-1} P_j^{(\mu-\frac{1}{2}, n-2j+\frac{1}{2}d-1)}(2\|x\|^2 - 1)\, Y_{\ell,n-2j}(x), \tag{2.5.5}$$

where

$$[h_{j,n}]^2 := \frac{(\mu + \frac{1}{2})_j (\frac{1}{2}d)_{n-j}}{j! (\mu + \frac{1}{2}(d+1))_{n-j}} \frac{(n - j + \mu + \frac{1}{2}(d-1))}{(n + \mu + \frac{1}{2}(d-1))}.$$

Then $\{P_{\ell,j} \mid 1 \le \ell \le r_{n-2j}^d,\ 0 \le j \le n/2\}$ is an orthonormal basis of $\mathcal{V}_n^d$.

Appell's biorthogonal polynomials These are two families of polynomials yielding bases $\{U_\alpha\}_{|\alpha|=n}$ and $\{V_\alpha\}_{|\alpha|=n}$ of $\mathcal{V}_n^d$, where the second basis is the monic basis, up to constant factors, and the first basis is biorthogonal to it.

(i) The family $\{U_\alpha\}$ is defined by the generating function

$$\Big((1 - \langle b, x\rangle)^2 + \|b\|^2(1 - \|x\|^2)\Big)^{-\mu} = \sum_{\alpha\in\mathbb{N}_0^d} b^\alpha U_\alpha(x), \qquad b \in \mathbb{R}^d,\ \|b\| < 1. \tag{2.5.6}$$

It satisfies the Rodrigues type formula,

$$U_\alpha(x) = \frac{(-1)^{|\alpha|} (2\mu)_{|\alpha|}}{2^{|\alpha|} (\mu + \frac{1}{2})_{|\alpha|} \alpha!} (1 - \|x\|^2)^{-\mu+\frac{1}{2}} \frac{\partial^{|\alpha|}}{\partial x^\alpha} (1 - \|x\|^2)^{|\alpha|+\mu-\frac{1}{2}}. \tag{2.5.7}$$

where $\frac{\partial^{|\alpha|}}{\partial x^\alpha} := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}$. Furthermore, it can be explicitly given as

$$U_\alpha(x) = \frac{(2\mu)_{|\alpha|}}{\alpha!} \sum_{\beta\le\alpha} \frac{(-1)^{|\beta|} (-\alpha)_{2\beta}}{2^{2|\beta|} \beta! (\mu + \frac{1}{2})_{|\beta|}} x^{\alpha-2\beta}(1 - \|x\|^2)^{|\beta|} \tag{2.5.8}$$

$$= \frac{(2\mu)_{|\alpha|}x^\alpha}{\alpha!} F_B\Big(-\tfrac{1}{2}\alpha, -\tfrac{1}{2}(\alpha-\mathbf{1}); \mu+\tfrac{1}{2}; x_1^{-2}(1-\|x\|^2), \dots, x_d^{-2}(1-\|x\|^2)\Big),$$

where $(-\alpha)_{2\beta} := (-\alpha_1)_{2\beta_1}\dots(-\alpha_d)_{2\beta_d}$, $\mathbf{1} := (1,\dots,1)$, $\beta \le \alpha$ means $\beta_i \le \alpha_i$ $(1 \le i \le d)$, and F_B is Lauricella's hypergeometric series of type B (see [34] and Chapter 3).

(ii) The family $\{V_\alpha\}$ is defined by the generating function

$$(1-2\langle b,x\rangle + \|b\|^2)^{-\mu-(d-1)/2} = \sum_{\alpha\in\mathbb{N}_0^d} b^\alpha V_\alpha(x), \qquad b\in\mathbb{R}^d,\ \|b\|<1. \tag{2.5.9}$$

The generating function implies that

$$\|b\|^n\, C_n^{\mu+\frac{1}{2}(d-1)}\left(\frac{\langle b,x\rangle}{\|b\|}\right) = \sum_{|\alpha|=n} b^\alpha V_\alpha(x). \tag{2.5.10}$$

The polynomial V_α can be written explicitly as

$$V_\alpha(x) = 2^{|\alpha|}x^\alpha \sum_{\gamma<\alpha} \frac{(\mu+\frac{1}{2}(d-1))_{|\alpha|-|\gamma|}\,(-\alpha+\gamma)_\gamma}{(\alpha-\gamma)!\,\gamma!}\, 2^{-2|\gamma|}x^{-2\gamma} \tag{2.5.11}$$

$$= \frac{2^{|\alpha|}(\mu+\frac{1}{2}(d-1))_{|\alpha|}}{\alpha!}\, x^\alpha F_B\Big(-\tfrac{1}{2}\alpha, -\tfrac{1}{2}(\alpha-\mathbf{1}); -|\alpha|-\mu-\tfrac{1}{2}(d-3); x_1^{-2},\dots,x_d^{-2}\Big).$$

Neither $\{U_\alpha\}_{|\alpha|=n}$ nor $\{V_\alpha\}_{|\alpha|=n}$ is an orthogonal basis of $\mathcal{V}_n^d$, but the two bases are biorthogonal to each other:

$$\int_{B^d} V_\alpha(x)U_\beta(x)W_\mu(x)\,dx = \frac{\mu+\frac{1}{2}(d-1)}{|\alpha|+\mu+\frac{1}{2}(d-1)}\,\frac{(2\mu)_{|\alpha|}}{\alpha!}\,\delta_{\alpha,\beta}. \tag{2.5.12}$$

The monic basis For each α, define

$$R_\alpha(x) := \frac{\alpha!}{2^{|\alpha|}(\mu+\frac{1}{2}(d-1))_{|\alpha|}}\, V_\alpha(x). \tag{2.5.13}$$

Then the polynomials R_α $(|\alpha|=n)$ form a monic basis of $\mathcal{V}_n^d$, i.e., $R_\alpha(x) = x^\alpha - Q_\alpha(x)$ with $Q_\alpha \in \Pi_{n-1}^d$. The $L^2(W_\mu, B^d)$ norm, $\|\cdot\|_{2,\mu}$, of R_α is the error of the best approximation of x^α, which satisfies a closed formula

$$\min_{P\in\Pi_{n-1}^d} \|x^\alpha - P(x)\|_{2,\mu}^2 = \|R_\alpha\|_{2,\mu}^2 = \frac{\lambda\,\alpha!}{2^{n-1}(\lambda)_n}\int_0^1 \prod_{i=1}^d P_{\alpha_i}(t)\, t^{n+2\lambda-1}\,dt, \tag{2.5.14}$$

where $\lambda = \mu + (d-1)/2 > 0$, $n = |\alpha|$, and P_{α_i} is the Legendre polynomial.

Reproducing kernel The kernel $P_n(\cdot,\cdot)$ of $\mathcal{V}_n^d$ with respect to W_μ, defined in (2.2.22), satisfies a compact formula. For $\mu>0$, $x,y\in B^d$,

$$P_n(x,y) = c_\mu \frac{2n+2\mu+d-1}{2\mu+d-1}\int_{-1}^1 C_n^{\mu+\frac{1}{2}(d-1)}\Big(\langle x,y\rangle + t\sqrt{1-\|x\|^2}\sqrt{1-\|y\|^2}\Big)(1-t^2)^{\mu-1}\,dt, \tag{2.5.15}$$

where $[c_\mu]^{-1} := \int_{-1}^{1}(1-t^2)^{\mu-1}dt$. For $\mu = 0$, $x, y \in B^d$ this degenerates to

$$P_n(x,y) = \frac{n+\frac{1}{2}(d-1)}{d-1} \sum_{\epsilon=0,1} C_n^{\frac{1}{2}(d-1)}\left(\langle x,y\rangle + (-1)^\epsilon \sqrt{1-\|x\|^2}\sqrt{1-\|y\|^2}\right). \qquad (2.5.16)$$

These formulas are essential for obtaining sharp results for convergence of orthogonal expansions. When $\mu = \frac{1}{2}(m-1)$, (2.5.15) can also be written as

$$P_n(x,y) = \frac{2n+m+d-2}{m+d-2} \int_{\mathbb{S}^{m-1}} C_n^{\frac{1}{2}(m+d)-1}\left(\langle x,y\rangle + \sqrt{1-\|x\|^2}\sqrt{1-\|y\|^2}\,\langle \xi, e_1\rangle\right) d\sigma_m(\xi), \qquad (2.5.17)$$

where $d\sigma_m$ is the normalized surface measure on $\mathbb{S}^{m-1}$.

For the constant weight $W_{1/2}(x) = (d+1)/\pi$, there is another formula for the reproducing kernel,

$$P_n(x,y) = (2nd^{-1}+1)\int_{\mathbb{S}^{d-1}} C_n^{d/2}(\langle x,\xi\rangle)\, C_n^{d/2}(\langle \xi,y\rangle)\, d\sigma_d(\xi). \qquad (2.5.18)$$

Further results and references For orthogonal bases on the ball, see [5, 28, 33]. There are further results on biorthogonal bases, see [5, 33]. For the monic basis, see [128]. Orthogonal bases consisting of ridge polynomials were discussed in [123], together with a Funk–Hecke type formula for orthogonal polynomials. The compact formulas (2.5.15) and (2.5.16) for the reproducing kernels were proved in [122] and used to study expansion problems, whereas the compact formula (2.5.17) was proved in [124, Theorem 2.6] (there take $H_2(\eta) = 1$). Formula (2.5.18) was proved in [91] in the context of approximation by ridge functions, and in [132] in connection with Radon transforms. In [7] three-term relations are used to develop an efficient numerical algorithm for the evaluation of orthogonal polynomials in (2.5.3) with $d = 2, 3$. For convergence and summability of orthogonal expansions, see §2.7.

2.5.2 Classical orthogonal polynomials on the simplex

For $x \in \mathbb{R}^d$, let $|x| := x_1 + \cdots + x_d$. Let $T^d := \{x \in \mathbb{R}^d \mid x_1, \ldots, x_d, 1-|x| \ge 0\}$ be the simplex in $\mathbb{R}^d$. The classical weight function on T^d is defined by

$$W_\kappa(x) := \frac{\Gamma(|\kappa| + \frac{1}{2}(d+1))}{\prod_{i=1}^{d+1}\Gamma(\kappa_i+\frac{1}{2})} x_1^{\kappa_1-\frac{1}{2}} \ldots x_d^{\kappa_d-\frac{1}{2}}(1-|x|)^{\kappa_{d+1}-\frac{1}{2}}, \quad \kappa_1,\ldots,\kappa_{d+1} > -\tfrac{1}{2}. \qquad (2.5.19)$$

Differential operator Orthogonal polynomials of degree n with respect to W_κ are eigenfunctions of a second order differential operator,

$$\left(\sum_{i=1}^{d} x_i(1-x_i)\frac{\partial^2}{\partial x_i^2} - 2\sum_{1\le i<j\le d} x_i x_j \frac{\partial^2}{\partial x_i \partial x_j} + \sum_{i=1}^{d}\left((\kappa_i+\tfrac{1}{2}) - (|\kappa| + \tfrac{1}{2}(d+1))x_i\right)\frac{\partial}{\partial x_i}\right)P$$
$$= -n\left(n+|\kappa|+\tfrac{1}{2}(d-1)\right)P, \qquad P \in \mathcal{V}_n^d, \quad (2.5.20)$$

where $|\kappa| = \kappa_1 + \cdots + \kappa_{d+1}$.

An orthonormal basis To state this basis we use the notation of $\mathbf{x}_j$ and α^j as in the first orthonormal basis on B^d of §2.5.1. We also put $\kappa^j := (\kappa_j, \ldots, \kappa_{d+1})$ $(j = 0, 1, \ldots, d+1)$ if $\kappa = (\kappa_1, \ldots, \kappa_{d+1})$. Then an orthonormal basis $\{P_\alpha\}_{|\alpha|=n}$ of $\mathcal{V}_n^d$ is given by

$$P_\alpha(x) := [h_\alpha]^{-1} \prod_{j=1}^{d} (1 - |\mathbf{x}_{j-1}|)^{\alpha_j} P_{\alpha_j}^{(a_j, \kappa_j - \frac{1}{2})} \left(\frac{2x_j}{1 - |\mathbf{x}_{j-1}|} - 1 \right), \tag{2.5.21}$$

where $a_j := 2|\alpha^{j+1}| + |\kappa^{j+1}| + \frac{1}{2}(d - j - 1)$ and h_α is given by

$$[h_\alpha]^2 := \prod_{j=1}^{d} \frac{(\kappa_j + \frac{1}{2})_{\alpha_j} \left(|\kappa^{j+1}| + \frac{1}{2}(d - j + 1)\right)_{|\alpha^j| + |\alpha^{j+1}|}}{\alpha_j! \left(|\kappa^j| + \frac{1}{2}(d - j + 2)\right)_{|\alpha^j| + |\alpha^{j+1}|}} \, \frac{2(a_j + \kappa_j + \alpha_j) + 1}{2(a_j + \kappa_j + 2\alpha_j) + 1}.$$

Appell's biorthogonal polynomials These are two families of polynomials yielding bases $\{U_\alpha\}_{|\alpha|=n}$ and $\{V_\alpha\}_{|\alpha|=n}$ of $\mathcal{V}_n^d$, where the second basis is the monic basis and the first basis is biorthogonal to it.

(i) The family $\{U_\alpha\}$ is defined by the Rodrigues type formula

$$U_\alpha(x) := x_1^{-\kappa_1 + \frac{1}{2}} \ldots x_d^{-\kappa_d + \frac{1}{2}} (1 - |x|)^{-\kappa_{d+1} + \frac{1}{2}} \frac{\partial^{|\alpha|}}{\partial x^\alpha} \left(x_1^{\alpha_1 + \kappa_1 - \frac{1}{2}} \ldots x_d^{\alpha_d + \kappa_d - \frac{1}{2}} (1 - |x|)^{|\alpha| + \kappa_{d+1} - \frac{1}{2}} \right). \tag{2.5.22}$$

(ii) The family $\{V_\alpha\}$ is explicitly defined by

$$\begin{aligned} V_\alpha(x) &:= \sum_{\beta \le \alpha} (-1)^{n + |\beta|} \left(\prod_{i=1}^{d} \binom{\alpha_i}{\beta_i} \frac{(\kappa_i + \frac{1}{2})_{\alpha_i}}{(\kappa_i + \frac{1}{2})_{\beta_i}} \right) \frac{(|\kappa| + \frac{1}{2}(d-1))_{n + |\beta|}}{(|\kappa| + \frac{1}{2}(d-1))_{n + |\alpha|}} \, x^\beta \\ &= \frac{(-1)^n (\kappa + \frac{1}{2})_\alpha}{(n + |\kappa| + \frac{1}{2}(d-1))_{|\alpha|}} \, F_A \left(n + |\kappa| + \tfrac{1}{2}(d-1), -\alpha; \kappa + \tfrac{1}{2}, x \right) \end{aligned} \tag{2.5.23}$$

where F_A denotes Lauricella's hypergeometric series of type A (see [34] and Chapter 3).

Neither $\{U_\alpha\}_{|\alpha|=n}$ nor $\{V_\alpha\}_{|\alpha|=n}$ is an orthogonal basis of $\mathcal{V}_n^d$, but the two bases are biorthogonal to each other:

$$\int_{T^d} V_\beta(x) U_\alpha(x) W_\kappa(x) \, dx = \frac{(\kappa + \frac{1}{2})_\alpha (\kappa_{d+1} + \frac{1}{2})_{|\alpha|}}{(|\kappa| + \frac{1}{2}(d+1))_{2|\alpha|}} \, \alpha! \, \delta_{\alpha,\beta}. \tag{2.5.24}$$

Monic orthogonal basis By (2.5.23) and (2.5.24) the polynomials V_α ($|\alpha| = n$) form a monic basis: $V_\alpha(x) = x^\alpha - Q_\alpha(x)$, $Q_\alpha \in \Pi_{n-1}^d$. Such polynomials can be defined more generally in view of the observation that the simplex T^d is associated with the permutation group of $X := (x_1, \ldots, x_d, x_{d+1})$, where $x_{d+1} = 1 - |x|$. For $x \in T^d$ and $\alpha \in \mathbb{N}_0^{d+1}$, define $X^\alpha := x_1^{\alpha_1} \ldots x_d^{\alpha_d} (1 - |x|)^{\alpha_{d+1}}$. For $|\alpha| = n$ let $R_\alpha(x) = X^\alpha - Q_\alpha(x)$ be the element of $\mathcal{V}_n^d$ such that Q_α is a polynomial of degree at most $|\alpha| - 1$. When $\alpha_{d+1} = 0$, $R_\alpha(x)$ agrees with $V_\alpha(x)$ in (2.5.23) up to a constant factor. The $L^2(W_\kappa, T^d)$ norm $\| \cdot \|_{2,\kappa}$ of R_α is the error of the best approximation of X^α, which satisfies a closed formula:

$$\min_{P\in\Pi^d_{n-1}} \|X^\alpha - P(x)\|^2_{2,\kappa} = \|V_\alpha\|^2_{2,\kappa} = \frac{(|\kappa|+\frac{1}{2}(d-1))(\kappa+\frac{1}{2})_\alpha}{(|\kappa|+\frac{1}{2}(d-1)_{2|\alpha|}} \int_0^1 \left(\prod_{i=1}^{d+1} P_{\alpha_i}^{(0,\kappa_i-\frac{1}{2})}(2t-1)\right) t^{|\alpha|+|\kappa|+\frac{d-3}{2}}\, dt. \tag{2.5.25}$$

If $\alpha_{d+1} = 0$, (2.5.25) gives the error of best approximation to x^α.

Reproducing kernel The kernel $P_n(\cdot,\cdot)$ of $\mathcal{V}^d_n$ with respect to W_κ, defined in (2.2.22), satisfies a compact formula. For $\kappa_i > 0$, $1 \le i \le d$, $x, y \in T^d$,

$$\begin{aligned} P_n(x,y) = c_\kappa \frac{2(2n+|\kappa|)+d-1}{2|\kappa|+d-1} \frac{(|\kappa|+\frac{1}{2}(d-1))_n}{(\frac{1}{2})_n} \\ \times \int_{[-1,1]^{d+1}} P_n^{(|\kappa|+(d-2)/2,-1/2)}\left(2z(x,y,t)^2-1\right)\left(\prod_{i=1}^{d+1}(1-t_i^2)^{\kappa_i-1}\right) dt, \end{aligned} \tag{2.5.26}$$

where $z(x,y,t) := \sqrt{x_1y_1}\, t_1 + \cdots + \sqrt{x_dy_d}\, t_d + \sqrt{1-|x|}\sqrt{1-|y|}\, t_{d+1}$ and

$$[c_\kappa]^{-1} := \int_{[-1,1]^{d+1}} \left(\prod_{i=1}^{d+1}(1-t_i^2)^{\kappa_i-1}\right) dt.$$

If some $\kappa_i = 0$, then the formula holds under the limit relation

$$\lim_{\lambda\to 0} c_\lambda \int_{-1}^1 g(t)(1-t)^{\lambda-1}\, dt = \tfrac{1}{2}(g(1)+g(-1)).$$

Further results and references For $d = 2$ the polynomials U_α on T^d were defined in [5] and the polynomials V_α were studied in [35]. For $d > 2$ they appeared in [45] when $W_\kappa(x) = 1$, and in [28] in general. The monic basis of polynomials R_α was studied in [128]. The formula (2.5.26) of the reproducing kernel appeared in [121]. A product formula for orthogonal polynomials on the simplex was established in [63]. The polynomials in (2.5.21) serve as generating functions for the Hahn polynomials in several variables [59, 136]. For convergence and summability of orthogonal expansions see §2.7.

2.5.3 Hermite polynomials of several variables

These are orthogonal polynomials with respect to the product weight

$$W_H(x) := \pi^{-d/2} e^{-\|x\|^2}, \qquad x \in \mathbb{R}^d. \tag{2.5.27}$$

Many properties are inherited from Hermite polynomials of one variable.

Differential operator Orthogonal polynomials of degree n with respect to W_H are eigenfunctions of a second order differential operator:

$$\left(\Delta - 2\sum_{i=1}^d x_i \frac{\partial}{\partial x_i}\right) P = -2nP, \qquad P \in \mathcal{V}^d_n. \tag{2.5.28}$$